Bounds for 3-AP-Free Sets (Kelley–Meka Bounds)
Chapter 2
Buka Meja Season 6

Romano Emmanuelle

Chiba University
Department of Mathematics and Computer Science

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Outline

Notational Conventions

- **Indicator Function.** $\mathbb{1}_*$ denotes an indicator function.
- **Finite Abelian Group.** G denotes any finite abelian group.
- **Normalized Expectation**[1] **(?).**

$$\mathop{\mathbb{E}}_{x\in G} f(x) := \frac{1}{|G|}\sum_{x\in G} f(x).$$

- **Inner product dan norm L^p.**
  For functions $f, g : G \to \mathbb{C}$, define

$$\langle f, g\rangle := \mathop{\mathbb{E}}_{x\in G} f(x)\overline{g(x)},$$

and

$$\|f\|_p := \left(\mathop{\mathbb{E}}_{x\in G} |f(x)|^p\right)^{1/p}, \qquad 1 \le p < \infty.$$

[1] namanya apa brok ini kwkwkw

Notational Conventions

- **Convolution and Difference Convolution.** The convolution and difference convolution[2] of f and g are defined by

$$f * g(x) := \mathop{\mathbb{E}}_{y \in G} f(y)\, g(x-y),$$

and

$$f \circ g(x) := \mathop{\mathbb{E}}_{y \in G} f(x+y)\, g(y),$$

respecetively. In particular, They satisfy

$$\langle f, g * h \rangle = \langle f \circ h, g \rangle.$$

[2] Convolution is commutative and associtative but difference convolution is neither commutative nor associative

Notational Conventions

- **(Relative)[3] Inner product and norm L^p.**
  Suppose $\mu : G \to \mathbb{R}_{\geq 0}$ s.t $\|\mu\|_1 = 1$. For functions $f, g : G \to \mathbb{C}$, define

$$\langle f, g \rangle_\mu := \mathop{\mathbb{E}}_{x \in G} \mu(x) f(x) \overline{g(x)},$$

  and

$$\|f\|_{p(\mu)} := \left(\mathop{\mathbb{E}}_{x \in G} \mu(x) |f(x)|^p \right)^{1/p}, \qquad 1 \leq p < \infty.$$

  We introduce some abuse of notation: $\|\mathbb{1}_A\|_{1(\mu)} = \mu(A)$.
- $\mu_A = \alpha^{-1} \mathbb{1}_A$, where $\alpha = |A|/|G|$.

[3] The special case above is the case when $\mu \equiv 1$.

Bird's-eye View of Chapter 2:

2: Unbalancing

Let $f : G \to \mathbb{R}$ satisfy $\widehat{f} \geq 0$. If $\|f\|_p \geq \frac{1}{4}$ for some $p \geq 1$, then there exists $p' \ll p$ such that $\|f + 1\|_{p'} \geq 1 + 1/8$.
In particular, if $\|\mu_A \circ \mu_A - 1\|_p \geq 1/4$, then $\|\mu_A \circ \mu_A\|_{p'} \geq 1 + 1/8$.

Lemma 7.

Let $\epsilon \in (0, 1)$ and $v : G \to \mathbb{R}_{\geq 0}$ satisfying $\|v\|_1 = 1$ and $\widehat{v} \geq 0$. Let $f : G \to \mathbb{R}$ such that

1. $\widehat{f} \geq 0$, or
2. $f = g \circ g$ and $v = h \circ h$ for some $g, h : G \to \mathbb{C}$.

If $\|f\|_{p(v)} \geq \epsilon$ for some $p \geq 1$, then $\|f + 1\|_{p'(v)} \geq 1 + \epsilon/2$ for some $p' \ll_\epsilon p$.

Bird's-eve View of Chapter 2: (2)

3. Dependent random choice (DRC)

If $\|\mu_A \circ \mu_A\|_p \geq 1 + 1/8$, then there exist subsets $A_1, A_2 \subset A$ of density at least $\alpha^{O(p)}$ such that $\langle \mu_{A_1} \circ \mu_{A_2}, \mathbb{1}_S \rangle \geq 1 - 1/32$, where

$$S := \{x \in G : \mu_A \circ \mu_A(x) > 1 + \tfrac{1}{16}\}.$$

Corollary 9.

Let $p \geq 1$ be n integer and $\epsilon > 0$. If $A \subseteq G$ is such that $||\mu_A \circ \mu_A|| \geq 1 + \epsilon$ and $S = \{x : \mu_A \circ \mu_A > 1 + \epsilon/2\}$, then there are $A_1, A_2 \subset G$, both of density $\gg \alpha^{2p+O_\epsilon(1)}$ such that $\langle \mu_{A_1} \circ \mu_{A_2}, \mathbb{1}_S \rangle \geq 1 - \epsilon/8$.

Bird's-eye View of Chapter 2: (3)

[4], To establish Corollary 9, we must follow a rather *long* path:

$$\text{Lemma 10} \implies \text{Lemma 8} \implies \text{Corollary 9} \rightsquigarrow 3.\ \text{DRC}$$

[4] "Unfortunately" (Zan'nendesuga)

Proof of Lemma 7

Now let us address the elephant in the room:

Lemma 7.

Let $\epsilon \in (0,1)$ and $v : G \to \mathbb{R}_{\geq 0}$ satisfying $\|v\|_1 = 1$ and $\widehat{v} \geq 0$. Let $f : G \to \mathbb{R}$ such that

1. $\widehat{f} \geq 0$, or
2. $f = g \circ g$ and $v = h \circ h$ for some $g, h : G \to \mathbb{C}$.

If $\|f\|_{p(v)} \geq \epsilon$ for some $p \geq 1$, then $\|f + 1\|_{p'(v)} \geq 1 + \epsilon/2$ for some $p' \ll_\epsilon p$.



Lemma 7

First, we need a claim:

Claim.

For any $\mathbb{Z} \ni k \geq 1$, $\langle v, f^k \rangle \geq 0$.

proof. Refer to the original paper. □

Proof of Lemma 7.

WLOG $p \geq 5$ is an odd integer. Since[5] $2\max(x, 0) = x + |x|$ for $x \in \mathbb{R}$:

$$\langle 2\max(f, 0), f^{p-1} \rangle_v = \langle v, f^p \rangle + \langle |f|, f^{p-1} \rangle_v \geq \|f\|_{p(v)}^p \geq \epsilon^p.$$

Consequently,

$$\mathbf{P} := \{x : f(x) \geq 0\} \Longrightarrow \langle \mathbb{1}_{\mathbf{P}}, f^p \rangle_v \geq \epsilon^p/2, \text{ and}$$
$$\mathbf{T} := \{x \in \mathbf{P} : f(x) \geq 3\epsilon/4\} \Longrightarrow \langle \mathbb{1}_{\mathbf{P}\setminus\mathbf{T}}, f^p \rangle_v \leq \epsilon^p/4.$$

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Proof of Lemma 7

Then, by C-S inequality:

$$v(\mathbf{T})^{1/2}||f||_{2p(v)}^{p} \geq \langle \mathbb{1}_{\mathbf{T}}, f^{p} \rangle_{v} \geq \epsilon^{p}/4. \tag{1}$$

On the other hand, by the triangle inequality:

$$||f||_{2p(v)} \leq 1 + ||f+1||_{2p(v)}.$$

Proof of Lemma 7

We consider the following cases:

a. $||f+1||_{2p(v)} > 2$. Taking $2p = p'$ and we are done.

b. $||f+1||_{2p(v)} \leq 2$. Then $||f||^p_{2p(v)} \leq 3$. Hence (**??**) becomes $v(T) \geq (\epsilon/10)^{2p}$. It follows that for any $p' \geq 1$:

$$||f+1||_{p'(v)} \geq \langle 1, |f+1|^{p'} \rangle_v^{1/p'} \geq \langle \mathbb{1}_{\mathbf{T}}, |f+1|^{p'} \rangle_v^{1/p'} \geq (1+3\epsilon/4)(\epsilon/10)^{2p/p}$$

For sufficiently large p' (depending on ϵ), we obtain the desired bound. □

Proof of Lemma 10

Lemma 10.

Let $p \geq 2$ be an even integer. Let $B_1, B_2 \subseteq G$ and $\mu := \mu_{B_1} \circ \mu_{B_2}$. For any finite set $A \subset G$ with density α and function $f : G \to \mathbb{R}_{\geq 0}$, there exist $A_1 \subseteq B_1$ and $A_2 \subseteq B_2$ such that

$$\langle \mu_{A_1} \circ \mu_{A_2}, f \rangle \leq 2 \frac{\langle (\mu_A \circ \mu_A)^p, f \rangle_\mu}{||\mu_A \circ \mu_A||^p_{p(\mu)}}$$

and

$$\min\left(\frac{|A_1|}{|B_1|}, \frac{|A_2|}{|B_2|}\right) \geq \frac{1}{4}\alpha^{2p}||\mu_A \circ \mu_A||^{2p}_{p(\mu)}.$$

The proof uses the same ideas as the original, but with a different ordering of the argument.

Lemma 10

The proof is rather long and exhausting. We proceed by divide the proof into 4 steps:

1. For $s \in G^p$, define $A_i(s) := B_i \cap \bigcap_{k=1}^{p} (A + s_k)$. We have

$$\langle (\mu_A \circ \mu_A)^p, f \rangle_\mu = \frac{\alpha^{-2p}|G|^2}{|B_1||B_2|} \mathop{\mathbb{E}}_{s \in G^p} \langle \mathbb{1}_{A_1(s)} \circ \mathbb{1}_{A_2(s)}, f \rangle.$$

In particular,

$$\|\mu_A \circ \mu_A\|_{p(\mu)}^p = \frac{\alpha^{-2p}\,|G|^2}{|B_1|\,|B_2|} \mathop{\mathbb{E}}_{s \in G} \alpha_1(s)\,\alpha_2(s).$$

Proof of Lemma 10

2. If $M = \frac{1}{2}\,\alpha^p \left(\frac{|B_1|\,|B_2|}{|G|^2}\right)^{1/2} \|\mu_A \circ \mu_A\|^p_{p(\mu)}$, then

$$\mathop{\mathbb{E}}_{s\in G^p}\left(\mathbb{1}_{\alpha_1(s)\alpha_2(s)<M^2}\,\alpha_1(s)\alpha_2(s)\right) < \frac{1}{2}\mathop{\mathbb{E}}_{s\in G^p}\left(\alpha_1(s)\alpha_2(s)\right).$$

3. $\alpha_1(s)\alpha_2(s) \geq M^2$.
4. Putting everything together.

Proof of Lemma 10: Step 1.

Mini-goal 1. (Mg1)

Understanding what is $A_i(s)$ and showing

$$(\mu_A \circ \mu_A)(b_1 - b_2)^p = \alpha^{-2p} \mathop{\mathbb{E}}_{s_1,\cdots,s_p \in G} \mathbb{1}_{\bigcap_{k=1}^p (A+s_k)}(b_1) \cdot \mathbb{1}_{\bigcap_{k=1}^p (A+s_k)}(b_2),$$

for $b_i \in B_i$, $i = 1, 2$.

Proof. (informal) First we carefully examine the case when $p = 1$:

$$\begin{aligned}(\mu_A \circ \mu_A)(b_1 - b_2) &= \mathop{\mathbb{E}}_{y \in G} \mu_A(b_1 - b_2 + y)\overline{\mu_A(y)} \\ &= \alpha^{-2} \mathop{\mathbb{E}}_{t \in G} \mathbb{1}_{A+t}(b_1) \cdot \mathbb{1}_{A+t}(b_2).\end{aligned}$$

Proof of Lemma 10: Step 1. (2)

Next, if $p = 2$,

$$(\mu_A \circ \mu_A)(b_1 - b_2)^2 = \alpha^{-4} \mathop{\mathbb{E}}_{t_1,t_2 \in G} \mathbb{1}_{A+t_1}(b_1)\mathbb{1}_{A+t_2}(b_2)\mathbb{1}_{A+t_1}(b_1)\mathbb{1}_{A+t_2}(b_2)$$
$$= \alpha^{-4} \mathop{\mathbb{E}}_{t_1,t_2 \in G} \mathbb{1}_{(A+t_1)\cap(A+t_2)}(b_1) \cdot \mathbb{1}_{(A+t_1)\cap(A+t_2)}(b_2).$$

Therefore, from our observation, we conclude that

$$(\mu_A \circ \mu_A)(b_1 - b_2)^p = \alpha^{-2p} \mathop{\mathbb{E}}_{s_1,\cdots,s_p \in G} \mathbb{1}_{\bigcap_{k=1}^p (A+s_k)}(b_1) \cdot \mathbb{1}_{\bigcap_{k=1}^p (A+s_k)}(b_2).$$

Now, for $i = 1, 2$, since $b_i \in B_i$, it is natural to define

$$A_i(s) := B_i \cap \bigcap_{k=1}^{p} (A + s_k),$$

where $s := (s_1, s_2, \ldots, s_p) \in G^p$. □

Proof of Lemma 10: Step 1. (3)

Next, we need one more mini-goal to examine the relative inner product $\langle (\mu_A \circ \mu_A)^p, f \rangle_\mu$:

Mini-goal 2. (Mg2)

$$\langle F^p, f \rangle_\mu = \frac{|G|}{|B_1||B_2|} \mathop{\mathbb{E}}_{b_1 \in B_1, b_2 \in B_2} F(b_1 - b_2)^p f(b_1 - b_2).$$

Proof. Exercise. □

Proof of Lemma 10: Step 1. (4)

Now, replacing $F = \mu_A \circ \mu_A$ yields:

$$\langle (\mu_A \circ \mu_A)^p, f \rangle_\mu \overset{Mg2}{=} \frac{|G|}{|B_1||B_2|} \mathop{\mathbb{E}}_{b_1 \in B_1,\, b_2 \in B_2} \big(\mu_A \circ \mu_A(b_1 - b_2)\big)^p f(b_1 - b_2)$$

$$\overset{Mg1}{=} \frac{|G|}{|B_1||B_2|} \mathop{\mathbb{E}}_{b_1 \in B_1,\, b_2 \in B_2} \alpha^{-2p} \mathop{\mathbb{E}}_{s \in G^p} \mathbb{1}_{A_2(s)}(b_1)\, \mathbb{1}_{A_2(s)}(b_2)\, f(b_1 - b_2)$$

$$\overset{Fubini}{=} \alpha^{-2p} \frac{|G|}{|B_1||B_2|} \mathop{\mathbb{E}}_{s \in G^p} \mathop{\mathbb{E}}_{b_1 \in B_1,\, b_2 \in B_2} \mathbb{1}_{A_2(s)}(b_1)\, \mathbb{1}_{A_2(s)}(b_2)\, f(b_1 - b_2)$$

$$= \alpha^{-2p} \frac{|G|^2}{|B_1||B_2|} \mathop{\mathbb{E}}_{s \in G^p} \left(\frac{1}{|G|} \mathop{\mathbb{E}}_{b_1 \in B_1,\, b_2 \in B_2} \mathbb{1}_{A_2(s)}(b_1)\, \mathbb{1}_{A_2(s)}(b_2)\, f(b_1 - b_2) \right)$$

$$\overset{Mg1}{=} \frac{\alpha^{-2p}|G|^2}{|B_1||B_2|} \mathop{\mathbb{E}}_{s \in G^p} \langle \mathbb{1}_{A_1(s)} \circ \mathbb{1}_{A_2(s)}, f \rangle.$$

Proof of Lemma 10: Step 1. (5)

In particular, when $f \equiv 1$, we have

$$\|\mu_A \circ \mu_A\|_{p(\mu)} = \frac{\alpha^{-2p}\,|G|^2}{|B_1|\,|B_2|} \mathop{\mathbb{E}}_{s \in G^p} \alpha_1(s)\,\alpha_2(s),$$

where $\alpha_i(s) = |A_i(s)|/|G|$.[6]

Proof. Exercise. (use Mini-goal 3) □

Mini-goal 3. (Mg3)

$$\langle F^p, 1 \rangle_\mu = ||F||^p_{p(\mu)}.$$

Proof. Exercise. □

[6] Note that $\alpha_i(s) = \mathbb{E}_{x \in G}\, \mathbb{1}_{A_i(s)}(x)$.

Proof of Lemma 10: Step 2.

We absolutely have forgotten what do we need to prove on step 2 so here it is:

Step 2.

If $M = \frac{1}{2}\alpha^p \left(\frac{|B_1||B_2|}{|G|^2}\right)^{1/2} \|\mu_A \circ \mu_A\|_{p(\mu)}^p$, then

$$\mathop{\mathbb{E}}_{s\in G^p}\left(\mathbb{1}_{\alpha_1(s)\alpha_2(s)<M^2}\,\alpha_1(s)\alpha_2(s)\right) \leq \frac{1}{2}\mathop{\mathbb{E}}_{s\in G^p}\left(\alpha_1(s)\alpha_2(s)\right).$$

Spoiler: The general idea is to use C-S inequality.

Proof of Lemma 10: Step 2. (2)

We first establish two mini-goals:

Mini-goal 4. (Mg4)

$$\mathop{\mathbb{E}}_{s\in G^p} \alpha_i(s) = \alpha^p \cdot |B_i|/|G|.$$

Proof. Exercise. (Use the fact that $\alpha_i(s) = \mathbb{E}_{x\in G}\, \mathbb{1}_{A_i(s)}(x)$) □

Mini-goal 5. (Mg5)

$$\mathop{\mathbb{E}}_{s\in G^p} \mathbb{1}_{\alpha_1\alpha_2<M^2}\alpha_1(s)\alpha_2(s) \leq M \mathop{\mathbb{E}}_{s\in G^p} \left(\alpha_1(s)^{1/2}\cdot \alpha_2(s)^{1/2}\right).$$

Proof. Exercise. (Use $\mathbb{1}_{\alpha_1\alpha_2<M^2}\alpha_1\alpha_2 \leq \mathbb{1}_{\alpha_1\alpha_2<M^2} M\sqrt{\alpha_1\alpha_2}$) □

Proof of Lemma 10: Step 2. (3)

Next,

$$\begin{aligned}
\frac{1}{2}\mathop{\mathbb{E}}_{s\in G^p}(\alpha_1(s)\alpha_2(s)) &\overset{\text{Step 1}}{=} \frac{1}{2}\frac{\alpha^{2p}|B_1||B_2|}{|G|^2}||\mu_A\circ\mu_A||^p_{p(\mu)} \\
&= \left(\alpha^p\frac{|B_1|}{|G|}\right)^{1/2}\left(\frac{\alpha^p|B_2|}{|G|}\right)^{1/2}\frac{\alpha^p}{2}\left(\frac{|B_1||B_2|}{|G|}\right)^{1/2}||\mu_A\circ\mu_A \\
&\overset{Mg4}{=} \left(\mathop{\mathbb{E}}_{s\in G^p}\alpha_1(s)\right)^{1/2}\left(\mathop{\mathbb{E}}_{s\in G^p}\alpha_2(s)\right)^{1/2}M \\
&\overset{C-S}{\geq} M\left(\mathop{\mathbb{E}}_{s\in G^p}\alpha_1(s)^{1/2}\alpha_2(s)^{1/2}\right) \\
&\overset{Mg5}{\geq} \mathop{\mathbb{E}}_{s\in G^p}\mathbb{1}_{\alpha_1\alpha_2<M^2}\alpha_1(s)\alpha_2(s),
\end{aligned}$$

as desired.

Proof of Lemma 10: Step 3.

Next, we are going to prove $\alpha_1(s)\alpha_2(s) \geq M^2$.

Mini-goal 6. (Mg6)

$$\mathop{\mathbb{E}}_{s\in G^p} \alpha_1(s)\alpha_2(s) \leq 2 \mathop{\mathbb{E}}_{s\in G^p} \mathbb{1}_{\alpha_1\alpha_2\geq M^2}\alpha_1(s)\alpha_2(s).$$

Proof. The result is obvious from step 2 and the fact that

$$\mathop{\mathbb{E}}_{s\in G^p} \alpha_1(s)\alpha_2(s) = \mathop{\mathbb{E}}_{s\in G^p} \mathbb{1}_{\alpha_1\alpha_2\geq M^2}\alpha_1(s)\alpha_2(s) + \mathop{\mathbb{E}}_{s\in G^p} \mathbb{1}_{\alpha_1\alpha_2< M^2}\alpha_1(s)\alpha_2(s).$$

Proof of Lemma 10: Step 3. (2)

It follows that

$$\mathop{\mathbb{E}}_{s\in G^p}\langle \mathbb{1}_{A_1(s)}\circ\mathbb{1}_{A_2(s)}, f\rangle \overset{\text{step 1}}{=} \underbrace{\frac{\langle(\mu_A\circ\mu_A)^p, f\rangle_\mu}{\|\mu_A\circ\mu_A\|_{p(\mu)}^p}}_{:=\eta} \mathop{\mathbb{E}}_{s\in G^p}\alpha_1(s)\alpha_2(s)$$

$$\overset{Mg6}{\leq} 2\eta \mathop{\mathbb{E}}_{s\in G^p}\mathbb{1}_{\alpha_1\alpha_2\geq M^2}\alpha_1(s)\alpha_2(s).$$

Therefore, there exists s such that

$$\langle \mathbb{1}_{A_1(s)}\circ\mathbb{1}_{A_2(s)}, f\rangle \leq 2\eta\mathbb{1}_{\alpha_1\alpha_2\geq M^2}\alpha_1(s)\alpha_2(s) \tag{2}$$

and hence $\alpha_1(s)\alpha_2(s)\geq M^2$. □

Exercise. The conclusion in red is not immediately obvious. Please provide a more detailed argument.

Proof of Lemma 10: Step 4.

Mini-goal 7. (Mg7)

$$\langle \mu_{A_1} \circ \mu_{A_2}, f \rangle = \frac{1}{\alpha_1(s)\alpha_2(s)} \langle \mathbb{1}_{A_1} \circ \mathbb{1}_{A_2}, f \rangle.$$

Finally,

$$\min\left(\frac{|A_1|}{|B_1|}, \frac{|A_2|}{|B_2|}\right) \geq \alpha_1(s)\alpha_2(s) \overset{\text{step 3}}{\geq} M^2 = \frac{1}{4}\alpha^{2p} ||\mu_A \circ \mu_A||_{p(\mu)}^{2p},$$

and

$$\langle \mu_{A_1} \circ \mu_{A_2}, f \rangle \overset{Mg7}{=} \frac{1}{\alpha_1(s)\alpha_2(s)} \langle \mathbb{1}_{A_1} \circ \mathbb{1}_{A_2}, f \rangle \overset{(??)}{\leq} 2\eta = 2\frac{\langle (\mu_A \circ \mu_A)^p, f \rangle_\mu}{\|\mu_A \circ \mu_A\|_{p(\mu)}^p}.$$

End of proof of Lemma 10. □

Lemma 8

Lemma 8.

Let $p \geq 1$ be an integer and $\epsilon, \delta > 0$. Let $B_1, B_2 \subseteq G$, and let $\mu = \mu_{B_1} \circ \mu_{B_2}$. For any finite set $A \subseteq G$ with density α, if

$$S := \{x \in G : \mu_A \circ \mu_A(x) > (1-\epsilon)||\mu_A \circ \mu_A||_{p(\mu)}\},$$

then there are $A_1 \subseteq B_1$ and $A_2 \subseteq B_2$ such that

$$\langle \mu_{A_1} \circ \mu_{A_2}, \mathbb{1}_S \rangle \geq 1 - \delta$$

and

$$\min\left(\frac{|A_1|}{|B_1|}, \frac{|A_2|}{|B_2|}\right) \gg \left(\alpha||\mu_A \circ \mu_A||_{p(\mu)}\right)^{2p+O_{\epsilon,\delta}(1)}.$$

Proof of Lemma 8.

TL;DR. Use Lemma 10 with $f \equiv \mathbb{1}_{G\setminus S}$.

Proof.

Choose an even integer $p \geq 1$ large enough such that $(1-\epsilon)^p \leq \delta/2$.

$$S := \{x \in G : \mu_A \circ \mu_A(x) > (1-\epsilon)||\mu_A \circ \mu_A||_{p(\mu)}\}.$$

For any $x \in G\setminus S$ we have

$$\mu_A \circ \mu_A(x) \leq (1-\epsilon)||\mu_A \circ \mu_A||_{p(\mu)}(x).$$

Then,

$$\begin{aligned}\langle \mu_A \circ \mu_A(x)^p, \mathbb{1}_{G\setminus S}\rangle_\mu &\leq \langle (1-\epsilon)^p||\mu_A \circ \mu_A||_{p(\mu)}^p, \mathbb{1}_{G\setminus S}\rangle_\mu \\ &= (1-\epsilon)^p||\mu_A \circ \mu_A||_{p(\mu)}^p,\end{aligned}$$

thus

$$\frac{\langle \mu_A \circ \mu_A(x)^p, \mathbb{1}_{G\setminus S}\rangle_\mu}{||\mu_A \circ \mu_A||_{p(\mu)}^p} \leq \delta/2. \tag{3}$$

Proof of Lemma 8. (2)

Therefore, it follows that

$$\langle \mu_{A_1} \circ \mu_{A_2}, \mathbb{1}_{G\setminus S} \rangle \overset{\text{Lemma 10}}{\leq} 2\frac{\langle (\mu_A \circ \mu_A)^p, \mathbb{1}_{G\setminus S} \rangle_\mu}{||\mu_A \circ \mu_A||^p_{p(\mu)}} \overset{(3)}{\leq} \delta. \tag{4}$$

On the other hand, since $\mathbb{1}_S = 1 - \mathbb{1}_{G\setminus S}$:

$$\begin{aligned} \langle \mu_{A_1} \circ \mu_{A_2}, \mathbb{1}_S \rangle &= 1 - \langle \mu_{A_1} \circ \mu_{A_2}, \mathbb{1}_{G\setminus S} \rangle \\ &\overset{(4)}{\geq} 1 - \delta, \end{aligned}$$

as desired. □

Corollary 9

Corollary 9, in particular, is the important special case of Lemma 8:

Corollary 9.

Let $p \geq 1$ be n integer and $\epsilon > 0$. If $A \subseteq G$ is such that

$$||\mu_A \circ \mu_A|| \geq 1 + \epsilon$$

and $S = \{x : \mu_A \circ \mu_A > 1 + \epsilon/2\}$, then there are $A_1, A_2 \subset G$, both of density

$$\gg \alpha^{2p+O_\epsilon(1)}$$

such that $\langle \mu_{A_1} \circ \mu_{A_2}, \mathbb{1}_S \rangle \geq 1 - \epsilon/8$.

Proof. Apply $B_1 = B_2 = G$, $\alpha = |A|/|G|$, $\delta = \epsilon/8$ to Lemma 8. □

Recap

What we have done:

$$\text{(K-M's assumption) } \langle \mu_A * \mu_A, \mu_C \rangle \leq 1/2$$

$$\text{(Step 1) } \overset{\text{Holder Lifting}}{\Longrightarrow} ||\mu_A \circ \mu_A - 1||_p \geq 1/4$$

$$\text{(Step 2) } \overset{\text{unbalancing}}{\Longrightarrow} ||\mu_A \circ \mu_A||_p \geq 1 + 1/8$$

$$\text{(Step 3) } \overset{\text{DRC}}{\Longrightarrow} \langle \mu_{A_1} \circ \mu_{A_2}, \mathbb{1}_S \rangle \geq 1 - 1/32.$$